

Task 25: Min Stack

Problem Description

Design a stack that supports push, pop, top, and retrieving the minimum element in constant time.

Java Solution

```
class MinStack {
    Stack<Integer> stack;
    Stack<Integer> minStack;

    public MinStack() {
        stack = new Stack<>();
        minStack = new Stack<>();
    }

    public void push(int val) {
        stack.push(val);
        if (minStack.isEmpty() || val <= minStack.peek()) {
            minStack.push(val);
        }
    }

    public void pop() {
        if (stack.pop().equals(minStack.peek())) {
            minStack.pop();
        }
    }

    public int top() {
        return stack.peek();
    }

    public int getMin() {
        return minStack.peek();
    }
}
```